

A background image showing a swarm of bees flying over a field of tall grass and yellow flowers. The bees are in various positions, some in flight and some on the flowers. The overall scene is a naturalistic representation of a bee colony.

Directing a Swarm of AI Agents for Fun and Profit

Adrian Cockcroft – Nov 17th, 2025
QConSF

Why did I choose this title?

Directing a Team of Agents - Management skill set

You aren't watching everything they do

You have to think and structure work like a director level manager

The agents build things you didn't ask for (or didn't know to ask for)

You have to make sure they remembered to run tests etc.

(Just like managing human dev teams)

But they do several days work in 15 minutes...



Fun and Profit

Coding with AI is fun!
Learn how to use the latest
tools earlier
Experiment on your own pet
projects
For a few \$\$, it's worth it to
help keep up



Code for work needs to be safe
Commercial deals take time
We need to be more careful and
responsible for paid work
*Speed of AI tool adoption is a
competitive advantage for your
workplace and career*

“The future is already here, it’s just unevenly distributed”

There is too much happening too fast

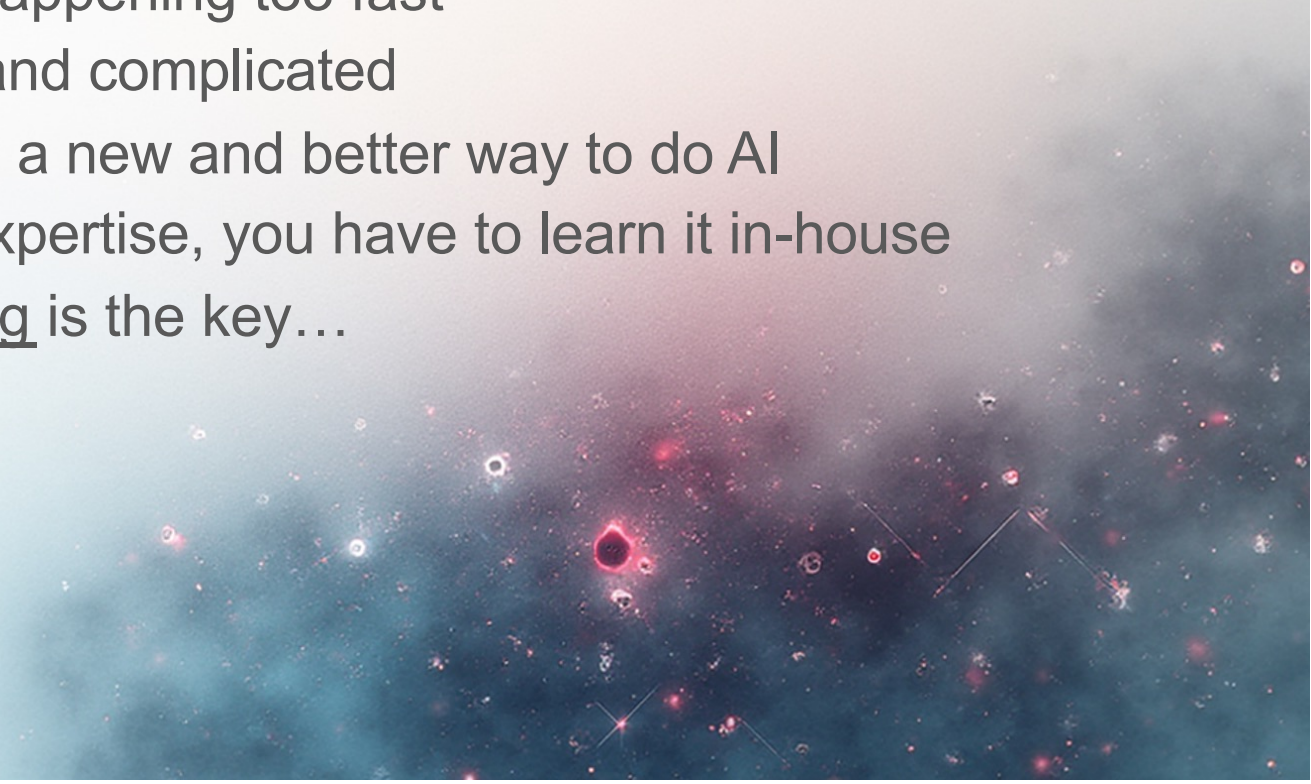
It will be confusing and complicated

Each month there is a new and better way to do AI

You can’t buy this expertise, you have to learn it in-house

Learning and sharing is the key...

Quote by William Gibson



Remember when computers weren't on demand?

Then:

Manager: I need a machine to run this code!

Ops: OK, I'll setup a meeting with the computer sales guy to figure out what we should buy, you need to get finance approval, get an order in, wait for delivery, then configure it. That should only take a few months... Then we'll depreciate it over 3 years.

Now:

```
ec2 = boto3.client("ec2")
ec2.run_instances(ImageID="ami-xxx",
                  InstanceType="t3.micro",
                  MinCount=1,
                  MaxCount=1)
```

Run it for a few minutes.

```
ec2.terminate_instances(InstanceID=...)
```

We called this **NoOps**, because we didn't need to talk to Ops (but some people got upset....)

What people do we (still) need to run code?

We need to run a cloud platform service using a DevOps/SRE team to implement the cloud architecture we want for our business.

Policy, security, guardrails, pre-built components, incident management etc.

Currently this is (relatively) slow changing, mature, well understood.



Remember when developers weren't on demand?

Then:

Manager: I need a dev to write this app!

HR: OK, I'll setup a meeting with the recruiters to figure out the hiring spec, get finance approval, post the job, interview applicants, then on-board them. That should only take a few months...

Manager: App is done, what do I do with this dev now?

Now:

```
$ npx claude-flow@alpha hive-mind spawn "write this app" --claude
```

Run for a few minutes until the code is written, then try it out and iterate.

We could call this **NoDevs**, because we don't need to hire developers to write code, but they might get upset...

What people do we (still) need to write code?

We need to run an agentic development service using a development platform team that implements the development architecture we want for our business.

Policy, security, guardrails, pre-built components, build testing, debugging etc.

Fast changing, chaotic, poorly understood.

Tooling and best practices that change every month...

Headcount moves from app development to platform development

Making it Work

The “easy fast path” code that just works...

Data capture and analysis in Python, automation that manipulates web pages and csv data

Example Python script Cursor session:

“There is a popup window with a downloadable CSV at <https://sustainability.aboutamazon.com/products-services/aws-cloud> which contains WUE and PUE and includes 2024 data. I would like a new python script code/aws-data-update.py to read this table from it's source URL, and update the table Cloud_Region_Metadata.csv with new data, saving it into a new csv that can be checked and uploaded. Use and update the existing code/requirements.txt for a python venv as needed”

Code ran first time, had no trouble getting the data out of AWS, I provided more specific guidance:

“Default the URL to the specific AWS URL mentioned, compare the data found to the existing data and only add new rows if new data is present, figure out the year from the URL and the existing file, no need to specify it”

“I ran it and saw it updating PUE data but overwriting WUE data with nan. The input data for regions that aren't in the format that contains two hyphens should be skipped as those are geographic rather than specific AWS region IDs.”

Code pushed, 185 lines of documentation, 636 lines of script, 527 loc. About 15 minutes elapsed time.... Fun!

<https://github.com/Green-Software-Foundation/real-time-cloud/blob/Dev/code/aws-data-update.py>



Adrian's Tips for better coding results – Use Python

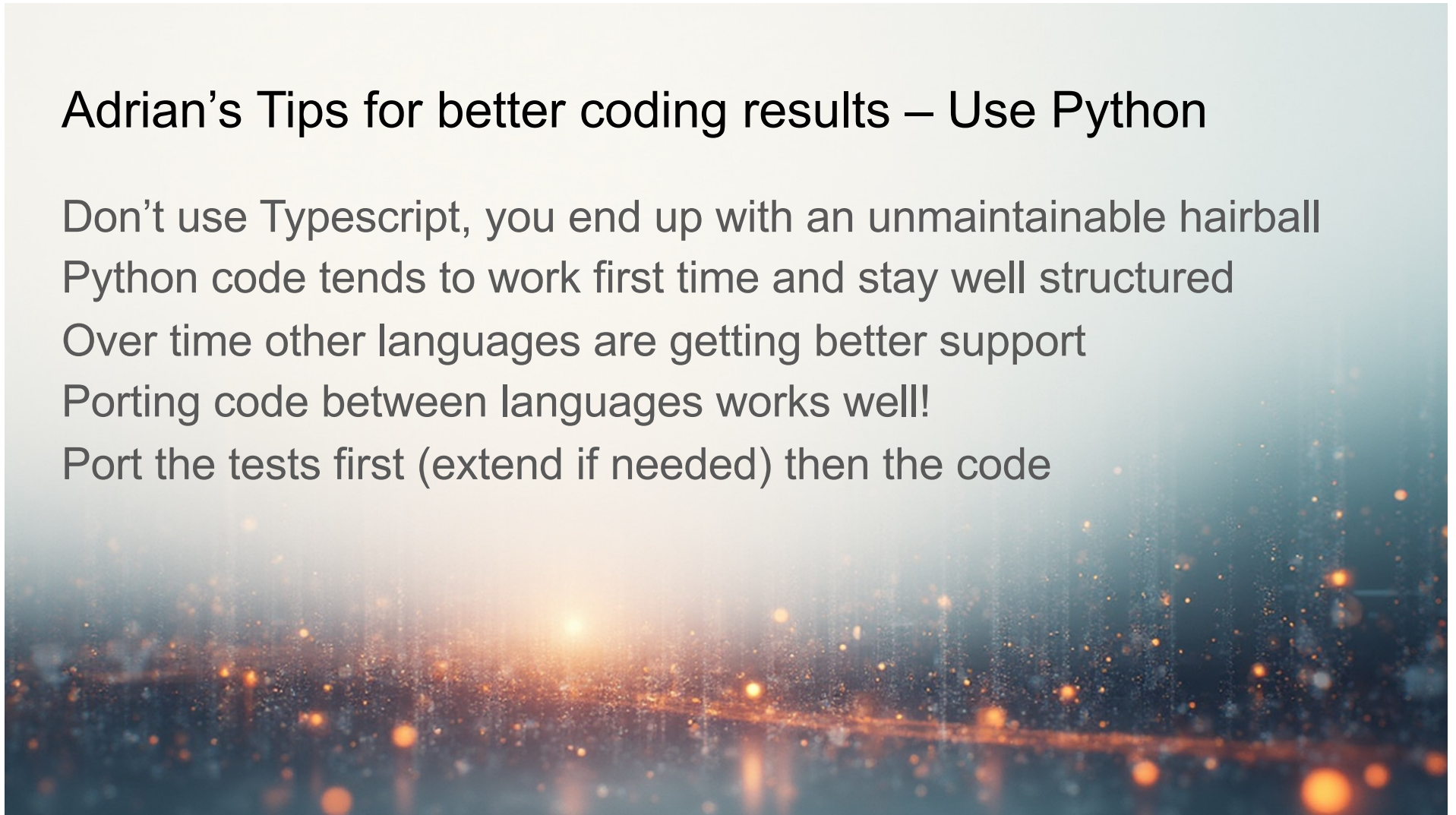
Don't use Typescript, you end up with an unmaintainable hairball

Python code tends to work first time and stay well structured

Over time other languages are getting better support

Porting code between languages works well!

Port the tests first (extend if needed) then the code



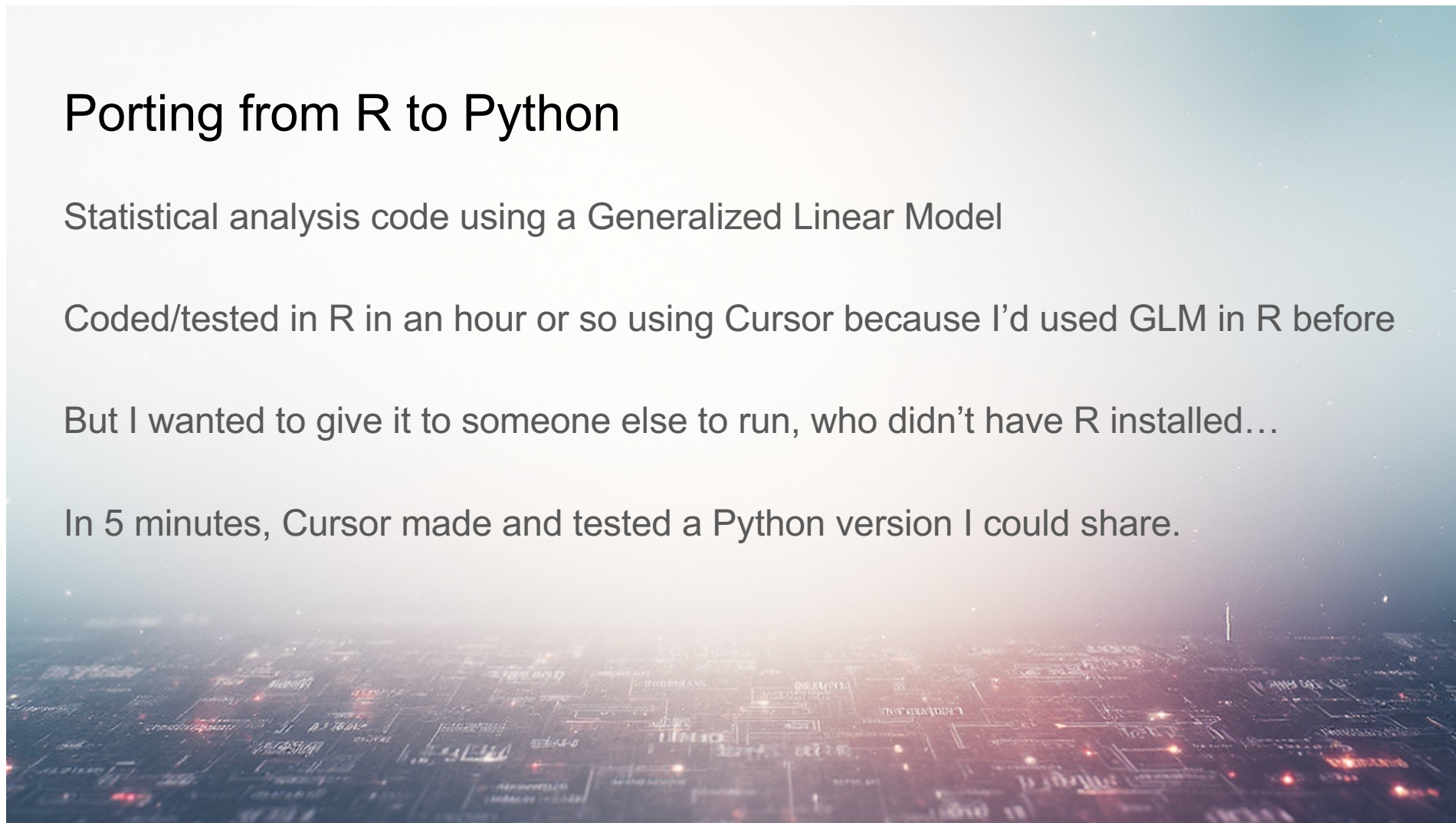
Porting from R to Python

Statistical analysis code using a Generalized Linear Model

Coded/tested in R in an hour or so using Cursor because I'd used GLM in R before

But I wanted to give it to someone else to run, who didn't have R installed...

In 5 minutes, Cursor made and tested a Python version I could share.



Techniques for better coding results – BDD Testing

Ask for a step by step plan, a UX guide and human testing instructions and check them before coding

Ask for Behavior Driven Development (BDD) tests

Given, When, Then structured in PyTest - check they make sense

Use the same G/W/T for mocked unit tests and integration tests



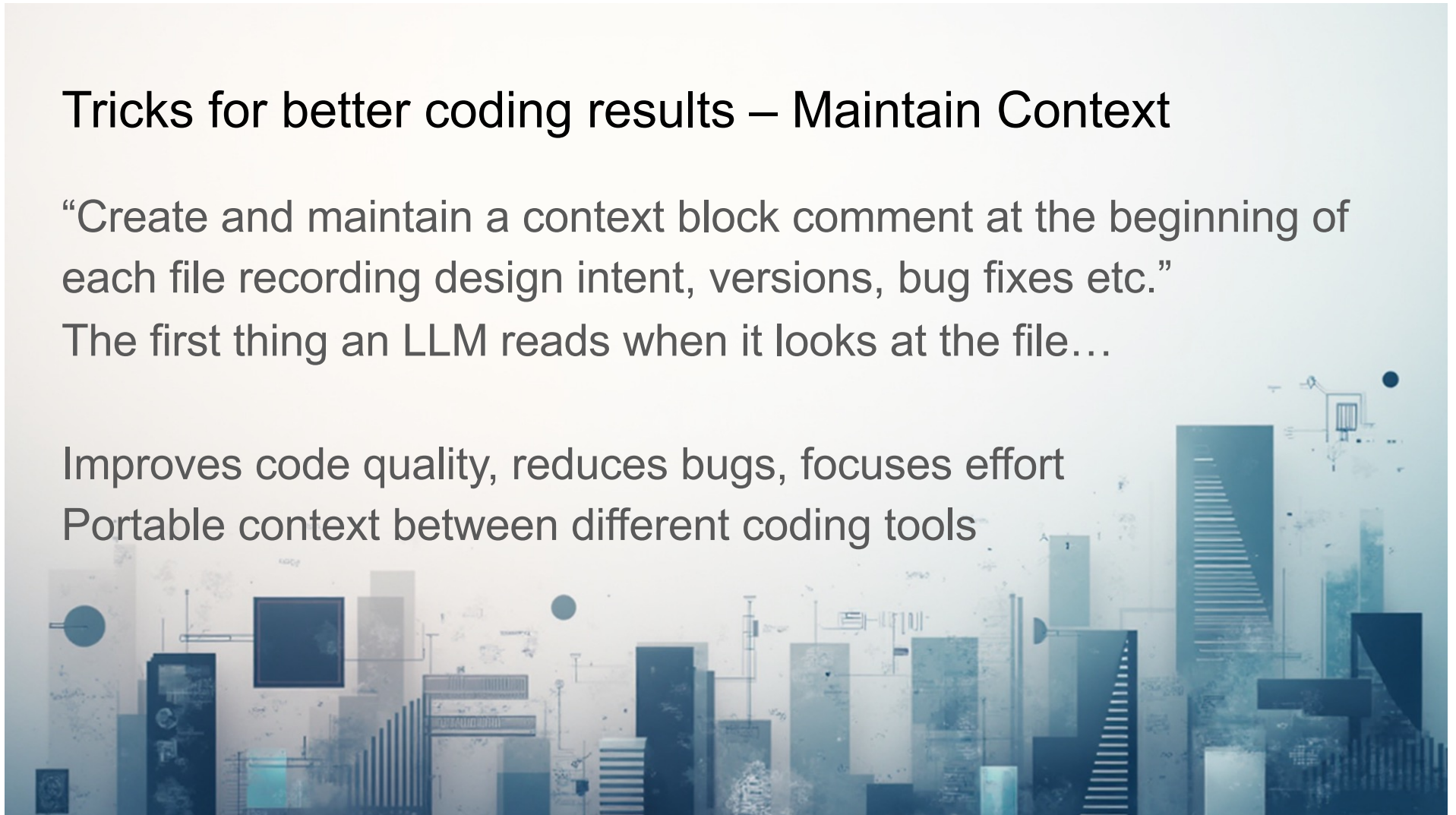
Tricks for better coding results – Maintain Context

“Create and maintain a context block comment at the beginning of each file recording design intent, versions, bug fixes etc.”

The first thing an LLM reads when it looks at the file...

Improves code quality, reduces bugs, focuses effort

Portable context between different coding tools



Tricks for better coding results – Tidy First

Clean up each phase of work before adding more features (Kent Beck)

Ask for a code review, refactor if needed

Make all compile and run time errors and warnings go away

Optimize slow code paths and add structured logging if appropriate

Update block comments, README and other docs, archive out of date docs



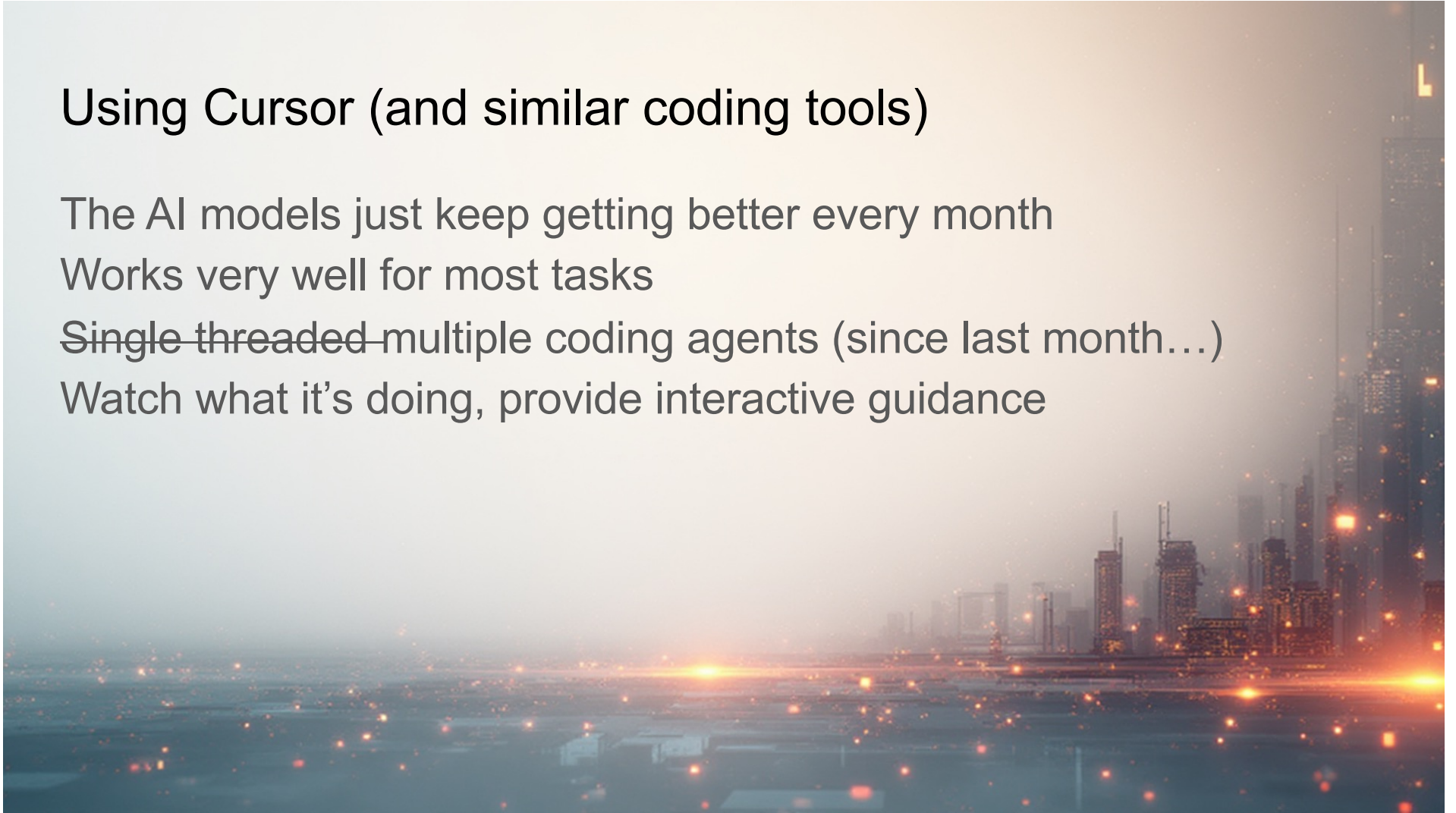
Using Cursor (and similar coding tools)

The AI models just keep getting better every month

Works very well for most tasks

~~Single threaded~~ multiple coding agents (since last month...)

Watch what it's doing, provide interactive guidance



A flexible power tool: Claude-Code

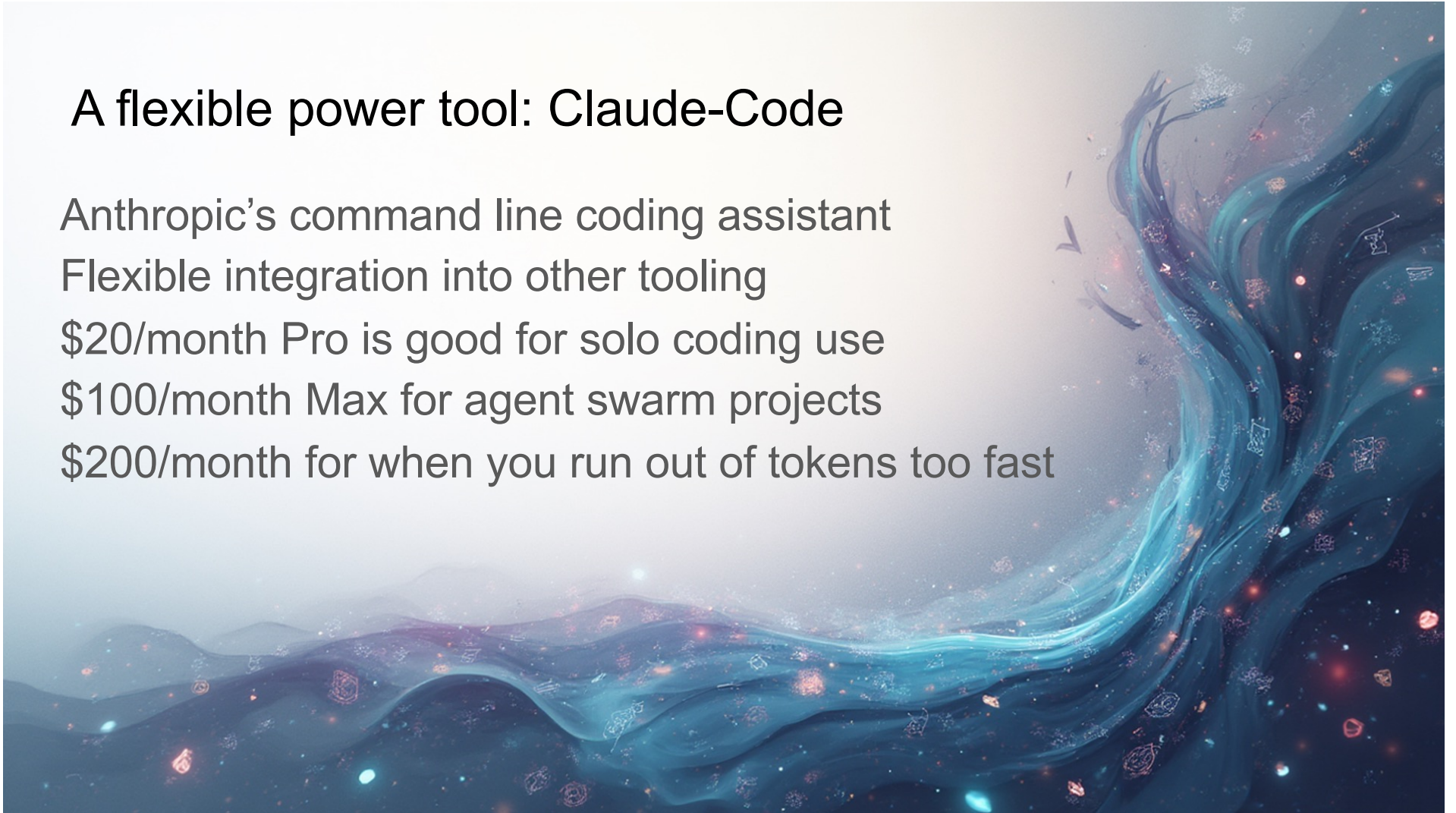
Anthropic's command line coding assistant

Flexible integration into other tooling

\$20/month Pro is good for solo coding use

\$100/month Max for agent swarm projects

\$200/month for when you run out of tokens too fast

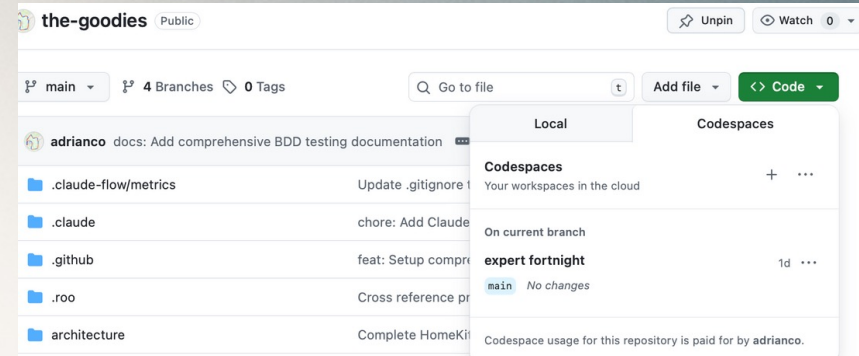


Github Codespaces

A safe place to run agents
based on Azure/Linux

All the agents can do is commit to the repo, so its safe to let them run:
`$ claude --dangerously-skip-permissions`

Monthly (\$20 worth) Codespaces free tier is plenty for fun projects
Safe container allows faster iteration and unattended agent work



Claude-Flow - Open Source by Reuven Cohen

Popular and powerful, but fast changing and too many options

Use to run an agent swarm in Github Codespaces

What is an Agent Swarm?

- 5-8 MCP specialized copies of Claude-code

- Architect, UX front end, Back end, Tester, DevOps etc.

- Communicating via ToDo lists and a shared memory service

Results are *significantly* better and faster for large projects

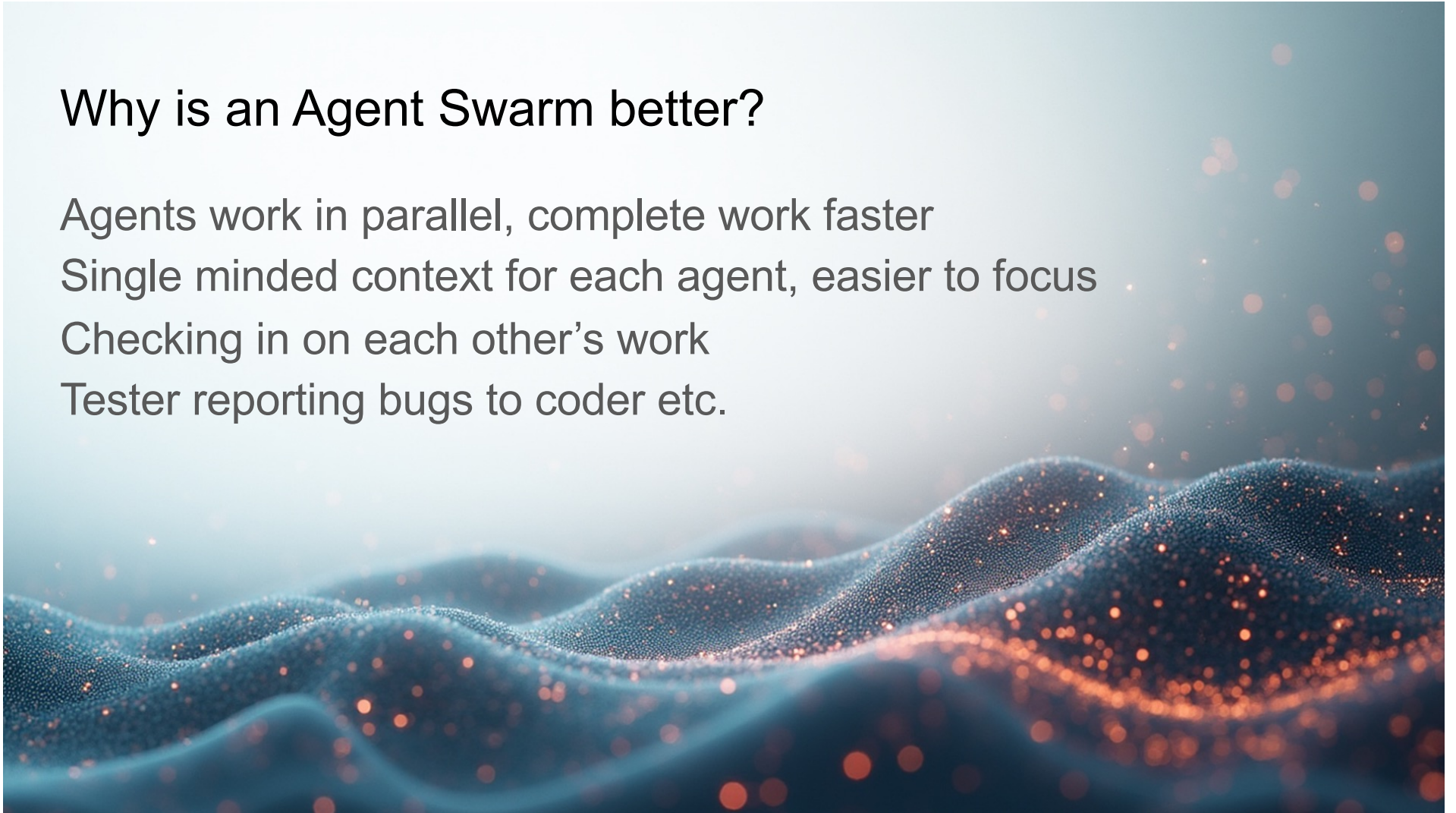
Why is an Agent Swarm better?

Agents work in parallel, complete work faster

Single minded context for each agent, easier to focus

Checking in on each other's work

Tester reporting bugs to coder etc.



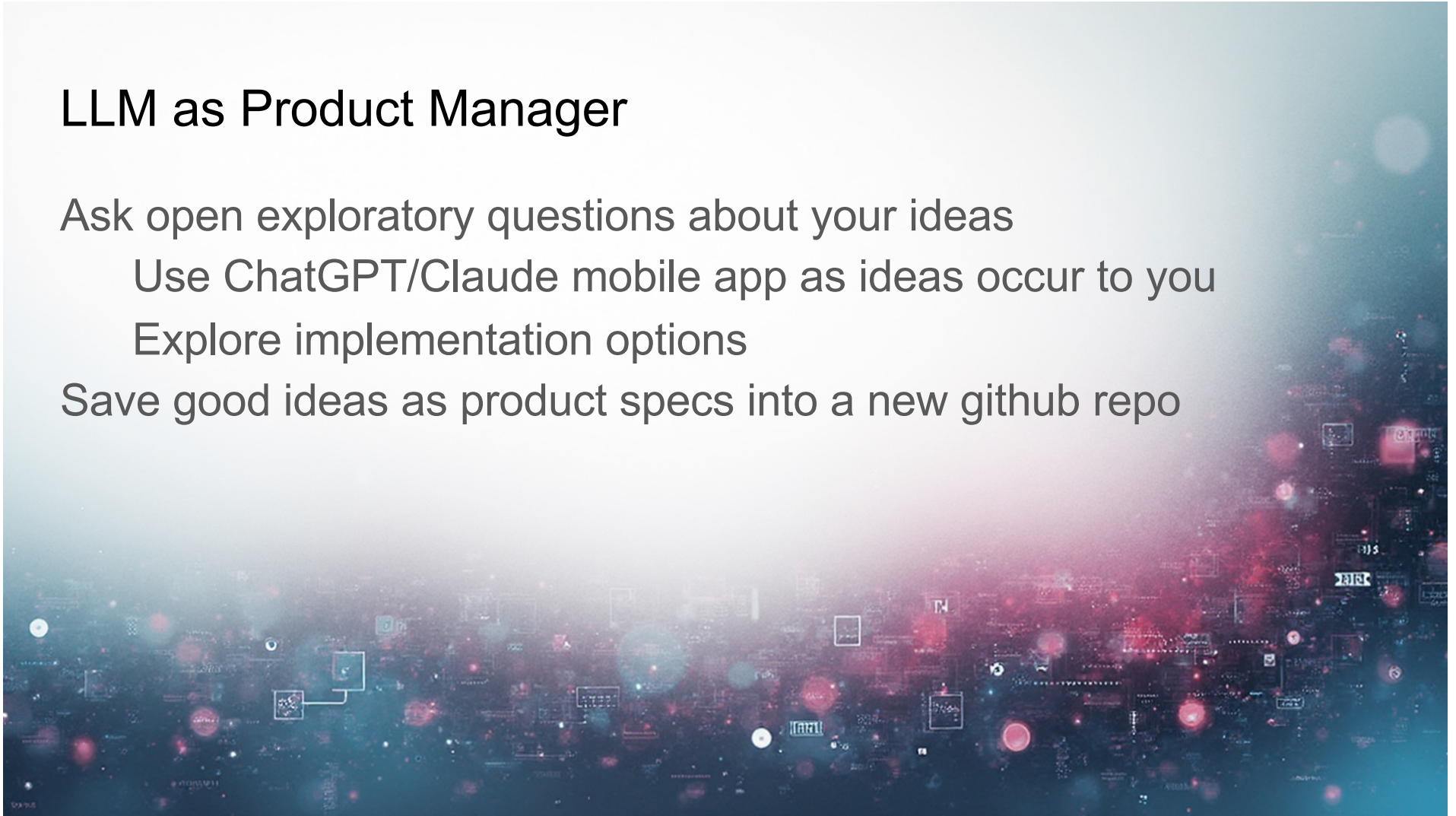
LLM as Product Manager

Ask open exploratory questions about your ideas

Use ChatGPT/Claude mobile app as ideas occur to you

Explore implementation options

Save good ideas as product specs into a new github repo



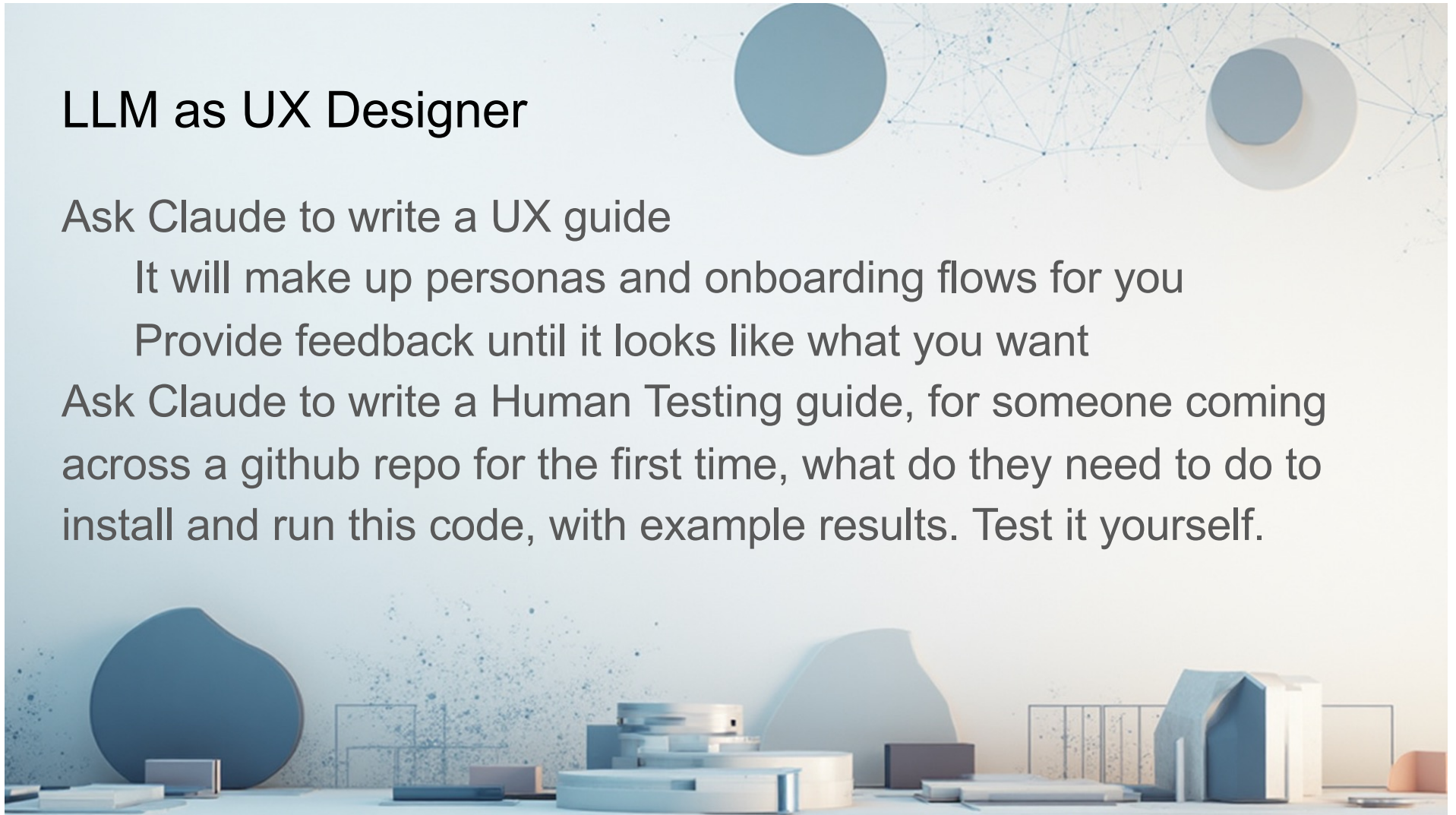
LLM as UX Designer

Ask Claude to write a UX guide

It will make up personas and onboarding flows for you

Provide feedback until it looks like what you want

Ask Claude to write a Human Testing guide, for someone coming across a github repo for the first time, what do they need to do to install and run this code, with example results. Test it yourself.



Tips for Directing an Agent Swarm

Tell them explicitly what you don't want them to do

“Write detailed step by step plans based on docs/idea.md, but don't write any code yet”

Then review the plans, point out issues and iterate until it's what you want

“Implement only the first step in the implementation plan, use comprehensive BDD tests to show it works, update all the plans and write a test results doc, then push the whole repo to github”

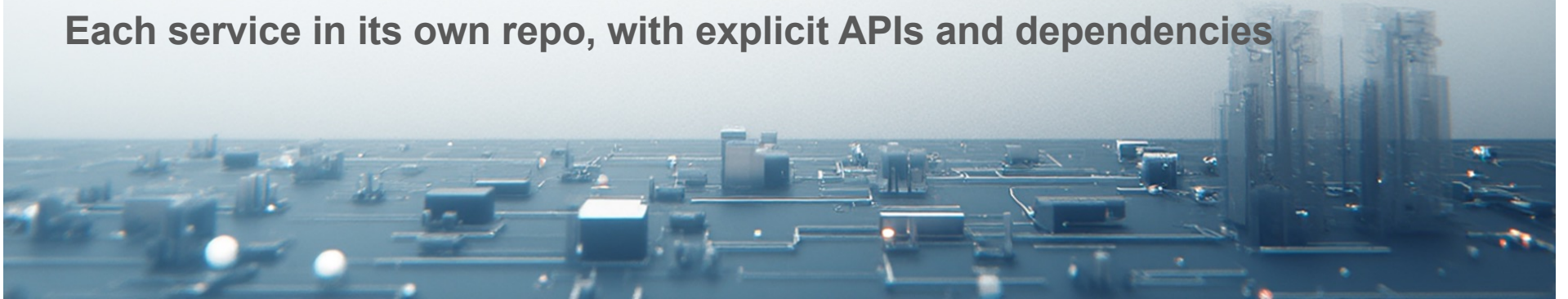
Microservices vs. Monoliths (redux)

The agents will build you a big messy “ball of mud” monolith If you let them...

Several people or agent teams will conflict working on the same code base

This is why we did Microservices in the first place, independently deliverable single function services. Easier to maintain developer context for changes.

Each service in its own repo, with explicit APIs and dependencies

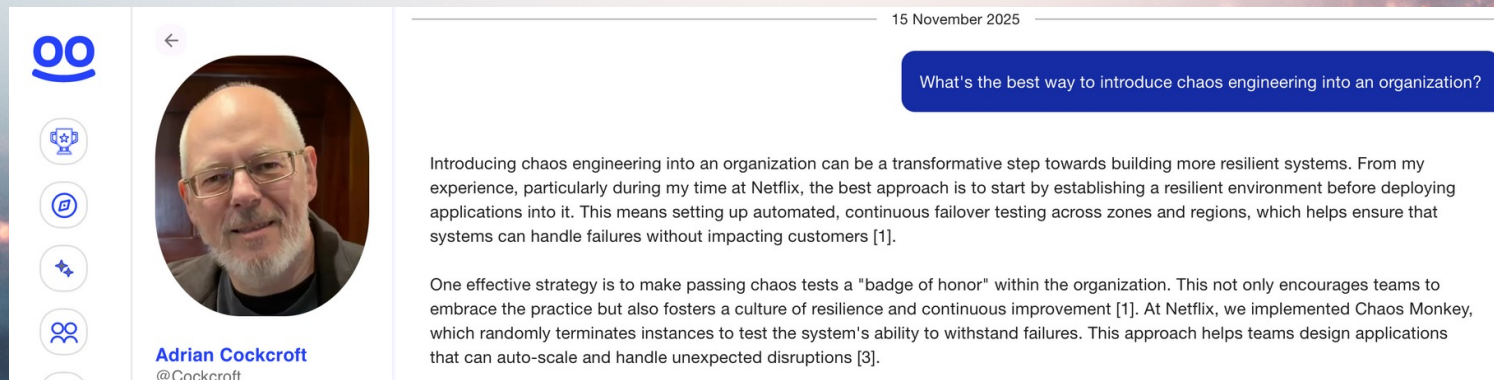


The “Hello World” of LLMs
is to build an MCP server

Persona Based LLM Developments

All your content, blog posts, podcasts, books, social media, in an LLM Persona as a Service - Soopra.ai - see <https://soopra.ai/Cockcroft>

Build it into an MCP server? <https://github.com/adrianco/megpt>
One of my early attempts at vibe coding...



Consciousness as an Observability Model

[Medium Post](#) by adrianco



Photo taken by Adrian at the Park Hyatt NingBo, China

Perhaps Consciousness is Just Human Observability?



adrian cockcroft · 6 min read · Jun 29, 2019

My definition of consciousness: **The things that go away when you are unconscious**

Conscious things can share their state
How are you? What are you thinking about?

We can use LLMs to make things conscious

My experiment is based on my own house
I want a knowledge graph MCP service as a mobile app that knows how to do stuff...

Consciousness Coding Projects

Reuven Cohen showed me...

Vibe Coding is so “Last Month...” — My First Agent Swarm Experience with claude-flow

adrian cockcroft · 20 min read · Jun 26, 2025

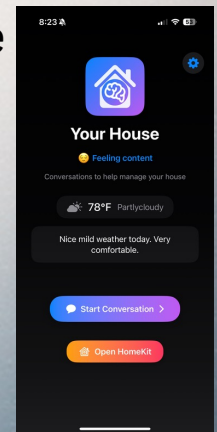
139 9



First attempt to use Claude-code
Python based Consciousness Service
Tried to build too much in one go
Ran but didn't work as I wanted

Second attempt to use Claude-code
Native Swift iOS app
Harder to get working than Python
Includes Weather, HomeKit, TTS
and voice recognition chat

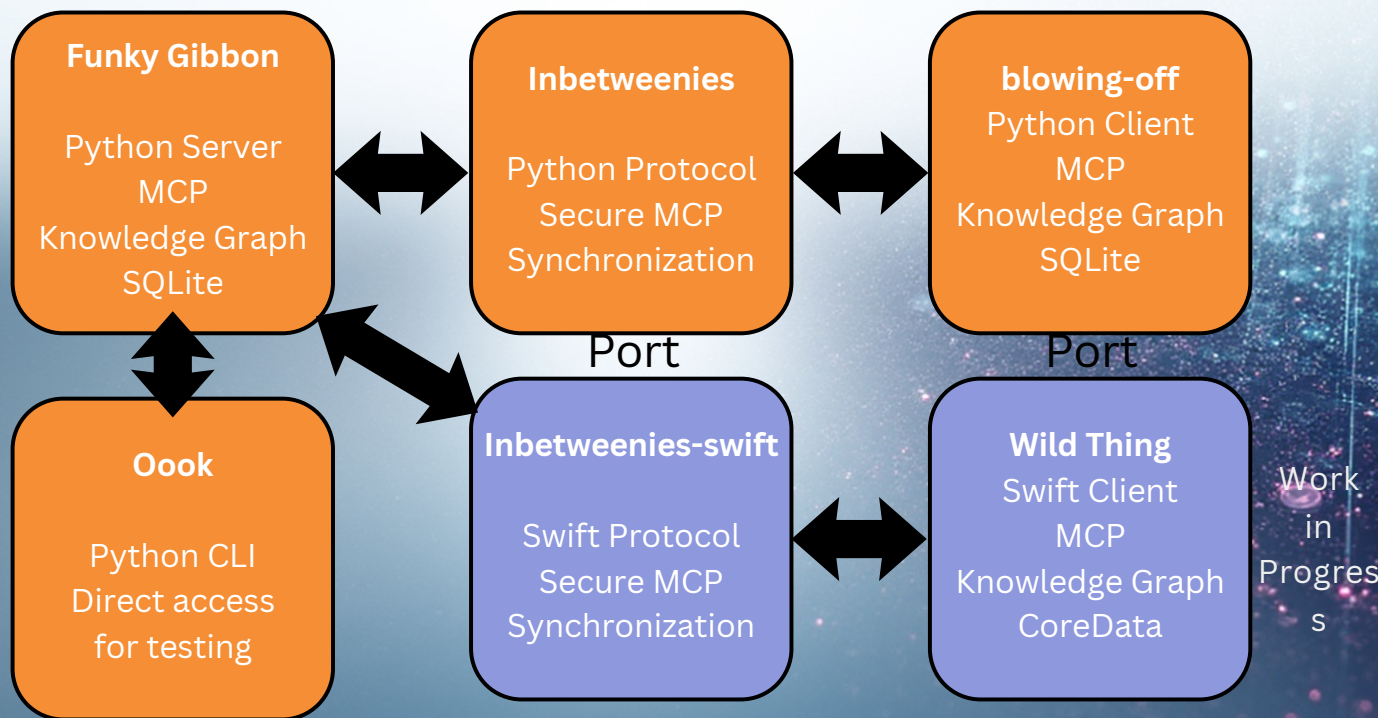
No backend as yet...



Distributed Portable MCP Service - The Goodies

<https://github.com/adrianco/the-goodies> - structured high quality clean code. All 225 tests pass.

Security: JWT authentication, rate limiting, audit logging. Access Control: Admin and guest role QR code access



Demo Agentic Coding at Nubank - This is all I said



Adrian: If I wanted to create an MCP knowledge graph for Brazilian soccer players, teams and games, is the data available to populate the graph?

Claude:... <too much output to show>

Adrian: don't need real time, need free access, this is for a demo, not commercial use

Claude:... <too much output to show>

Adrian: I will want to connect the MCP server to Claude and have a set of sample questions for the demo that Claude looks up via MCP to come up with the answers

Claude:... <too much output to show>

Adrian: save this entire conversation as a document that I can implement elsewhere



Test Questions from the doc Claude wrote

Simple Lookups:

1. "Who scored the most goals for Flamengo in 2023?"
2. "What teams has Neymar played for in his career?"
3. "When was the last match between Palmeiras and São Paulo?"

Relationship Queries:

1. "Which players have played for both Corinthians and Palmeiras?" (rival teams)
2. "Show me all Brazilian players who moved to European clubs in 2024"
3. "What teams did Ronaldo Nazário play for during his career?"

Aggregate/Analysis:

1. "Which team has won the most Brasileirão titles?"
2. "Who are the top 5 goal scorers in Brazilian Serie A history?"
3. "What's Flamengo's win rate against Internacional in the last 10 years?"

Complex Multi-hop:

1. "Find players who scored in a Copa do Brasil final and later played in Europe"
2. "Which coaches have managed multiple championship-winning teams?"
3. "Show me teammates of Neymar at Santos who also made it to European leagues"



Pre-Work

Created a new public Github Repo

<https://github.com/adrianco/demo-brazil>

Started a Codespace

Installed Claude-Code and Claude-Flow

Copied the document to the repo

Installed Neo4j

Now it's YOLO-demo-Time!



I ran this command to see what happened

```
$ npx claude-flow@alpha swarm "Read brazilian-soccer-mcp-guide.md and implement phases 1,2 and 3 as described and test using BDD GWT structured PyTest. Use Neo4j as documented in docs/neo4j-setup.md, maintain a detailed context block comment at the start of every code file. Finally update README.md to describe what was done and push everything to github" --claude
```

It took about an hour elapsed time, then I tinkered with it for another hour

Look at <https://github.com/adrianco/demo-brazil> to see the setup and results



Swarm to-do list

≡ prompts.txt

```
342
343 * Initializing swarm coordination for Brazilian Soccer MCP... (esc to interrupt · ctrl+t to show todos)
344 |   Next: Read and analyze Phase 1 requirements from guide
345
346   □ Initialize swarm coordination for Brazilian Soccer MCP
347   □ Read and analyze Phase 1 requirements from guide
348   □ Read and analyze Phase 2 requirements from guide
349   □ Read and analyze Phase 3 requirements from guide
350   □ Set up project structure and directories
351   □ Create Neo4j connection module
352   □ Implement data loader for Kaggle datasets
353   □ Create graph schema and entities
354   □ Implement MCP server with core tools
355   □ Write BDD tests using pytest-bdd
356   □ Test Phase 1 implementation
357   □ Test Phase 2 implementation
358   □ Test Phase 3 implementation
359   □ Update README.md with documentation
360   □ Commit and push to GitHub
361
```

Brazilian Soccer MCP Benchmark

Run a swarm with the same objective to see how the swarm results evolve...

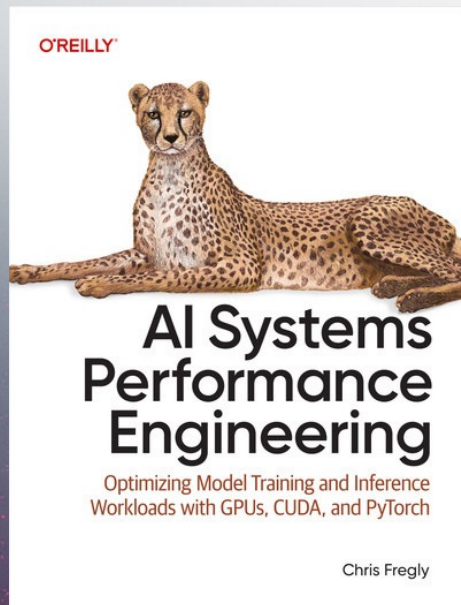
First try: Sept 30th claude-code swarm [adrianco/demo-brazil](#)

Second try: Oct 30th claude-code hive-mind [adrianco/brazil-bench-hive](#)

Hive results took less time to complete and look better



Re-coding a big project from TypeScript to Python



I was a reviewer of this excellent book by Chris Fregly that is released this month.

A startup I'm working with wanted a demo version of their app idea, and Chris built it over a week or two using Cursor, mostly in TypeScript. I provided a MCP knowledge graph component in Python.

I later used claude-flow hive-mind to rewrite it all in Python starting with the BDD tests, using the tests to drive the rewrite to completion.

What's Missing?

I'm doing too many repetitive mindless management tasks

“Make 100% of the tests pass”, “Tidy first”,

“Yes, keep going until you have finished”, “Archive old docs”,

“No don't skip that bit that didn't work”, “Push changes to github”,

“Compare the Python code functionality with the Typescript code again”

Need a Director agent. The Hive Queen is more of a line manager

Really, need a complete systemic platform for managing agent development

Break down tasks, manage to guardrails and policies

Automating the need for Development Management away... (perhaps **NoMan?**)

One more thing - Developer Carbon Footprint?

Based on estimates and emissions factors that seem plausible to me:

U.S. software developer carbon footprint around 20 tons CO₂e/yr

- Work-attributable ops only (commute, office power, work travel, devices): ~0.5–1.5 kg CO₂e/h
- Full personal footprint allocated to work hours ($\approx 20 \text{ tCO}_2\text{e/yr} \div \sim 2,000 \text{ h}$): ~10 kg CO₂e/h
- Human developers work at less than 1000 tokens/hr

Claude's carbon footprint per million tokens around 0.20 kg CO₂e

- Claude runs at around a million tokens/hr
- Assuming 10 Claude tokens per human token due to extra re-work, still 100x human
- **Carbon footprint of a US based human developer may be 5000 times higher than Claude?**

[Link to shared ChatGPT discussion session](#) if you want to change assumptions

References

<https://www.anthropic.com/claude-code>

<https://github.com/ruvnet/claude-flow>

Adrian's swarm coding experiments since June 22nd, 2025 - making my house self-aware

<https://github.com/adrianco/consciousness> - first try, did too much with claude-flow in one go

<https://github.com/adrianco/c11s-house-ios> - built a Swift/iOS app from scratch

<https://github.com/adrianco/the-goodies> - distributed MCP knowledge graph in Python

Brazilian soccer

<https://github.com/adrianco/demo-brazil> - the original test with claude-code swarm

<https://github.com/adrianco/brazil-bench-hive> - same starting point with claude-code hive-mind

<https://adrianco.medium.com> - Blog posts

<https://linkedin.com/in/adriancockcroft> - Discussion about agentic coding



YMMV

Questions?